Exercise 5 - Pointers and References

This weeks exercise will introduce you to pointers and references, as well as revisiting structs.

If you have any questions or problems, please write in the Moodle!

Exercise 1 — Pointers vs. References

For this Exercise write your functions into `main.cpp`.
Try out your code as usual in `main.cpp`.

a) Write the function `increment(int* address_to_number)`, incrementing the number at the given address. Analogous to this, write the function `increment(int& number)` doing exactly the same. *Which function is easier to write and understand in your opinion?*

b) Declare the struct `Element` storing an integer `value` as well as a `pointer` to another Element `next`. Than create two `Elements` `first` and `second`. `First` should store the `value 1` and point to `nothing` while `second` should store the `value 2` and point to `first`.
Print the `value` *from* `first` *through* `second`*'s* `pointer` *to* `first`.

Exercise 2 — References

For this Exercise write your structs and methods into `main.cpp`.
Try out your code as usual in `main.cpp`.

a) Declare the struct `Teacher` having a `std::string` as `name` as well as a `bool` field `isBooked` which should be `false` by default. Similarly declare the struct `Student` also having a `name` as well as a field called `classTeacher` which is a `reference` to a `Teacher`.

b) Create a `Teacher` you remember teaching your class from school. Also create yourself and your best friend back from school as `Students` referencing to that teacher. Write a method `print_student_and_teacher()`, which should print the following output:
`<Name>s teacher is <Teacher_Name>.` i.E. `Pauls teacher is Mrs. Flower.`

c) For the `struct Student` write a method `bookAppointment()` which checks, if the `classTeacher` `isBooked` or not and if not, books an appointment with them. Finally write the method `appointAppointment()` for the `struct Teacher` setting `isBooked` to `false` again.

Make sure both methods print appropriate messages to the console, depending on the state of `isBooked`.